Paleomagnetism of the Franklin Diabases

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Diabase intrusions, which have a probable mean age of 675 m.y., occur in a giant arc across the northern part of the Canadian Shield. They have a paleomagnetic pole position of 167° E, 8° N ($\alpha_{95} = 5°$), and at least one reversal. The paleomagnetic results indicate that the northern Canadian Shield was near the equator at the time of intrusion, and this is in agreement with evidence of warm conditions of deposition found in erosional remnants of related late Proterozoic sediments. Some of the diabase, near the shores of Davis Strait, may have been partially remagnetized during the opening of Baffin Bay in the early Tertiary.

Introduction

In the Phanerozoic, episodes of igneous activity have produced enormous arrays of basic extrusive and hypabyssal rocks which are characterized by internal uniformity of K–Ar ages and by uniformity of paleomagnetic directions; for example, the Tasmanian dolerite of Australia (McDougal 1961; Irving 1963), the Ferrar dolerite of Antarctica (McDougall 1963; Bull *et al.* 1962), and the Karroo dolerite and Drakensburg Volcanics of southern Africa (Graham and Hales 1957; McDougall 1963; van Zijl *et al.* 1962; McElhinny and Jones 1965). The Canadian Shield contains similar widespread diabase (dolerite) and related extrusive rocks of many different ages, and if these can be arranged into a series of discrete igneous episodes they will provide chronostratigraphic markers of some importance. As in the case of Phanerozoic basic intrusions, paleomagnetic and radioisotopic techniques should be powerful tools in correlating the Precambrian basic intrusions that lie within blocks that have not subsequently been internally deformed. Therefore, as part of a general investigation of the diabase of the Canadian Shield, the paleomagnetism of some diabases from northern Canada has been studied, and is described in this paper. The evidence for the K–Ar age of these intrusives is also reviewed. These intrusions are widespread in the district of Franklin and so they are referred to here (informally) as the Franklin diabases. For convenience of description the Franklin diabases are arranged into the following three groups.

(1) The Coronation and Bathurst Intrusions

A series of intrusions forms a belt near Bathurst Inlet where it cuts Archean and Proterozoic rocks (Fig. 1). The intrusions do not cut a sandstone unit that Fraser (1963) has correlated with late Cambrian strata on the south side of Victoria Island (Thorsteinsson and Tozer 1962). Intrusive diabase sheets also occur in a 1000 m sequence of gently dipping Hadrynian sediments that overlies the Coppermine River Volcanics near Coronation Gulf. The Bathurst and Coronation intrusives are almost certainly coeval.

(2) Boothia and Other Dikes

This is a miscellaneous group of NE-trending dikes near Hardisty Lake south of Great Bear Lake (Lord 1952; Fraser 1967), and NW-trending dikes on Boothia and Melville peninsulas, and in New Quebec (Fig. 2). Individual dikes range up to 200 m in thickness and can be traced for more than 100 km.

(3) Baffin Dikes

On Baffin Island (Fig. 2) there is a spectacular swarm of NW-trending dikes. Individual bodies range up to 200 m in width and can be traced for more than 100 km.

Age Determinations

The potassium–argon (K–Ar) age determinations made by the Geological Survey of Canada are listed in Table 1. Leech (1966) has also reported determinations on a NNE striking dike (a few miles south of site 6, Fig. 1) which apparently belongs to the Coronation and Bathurst intrusions; ages ranging

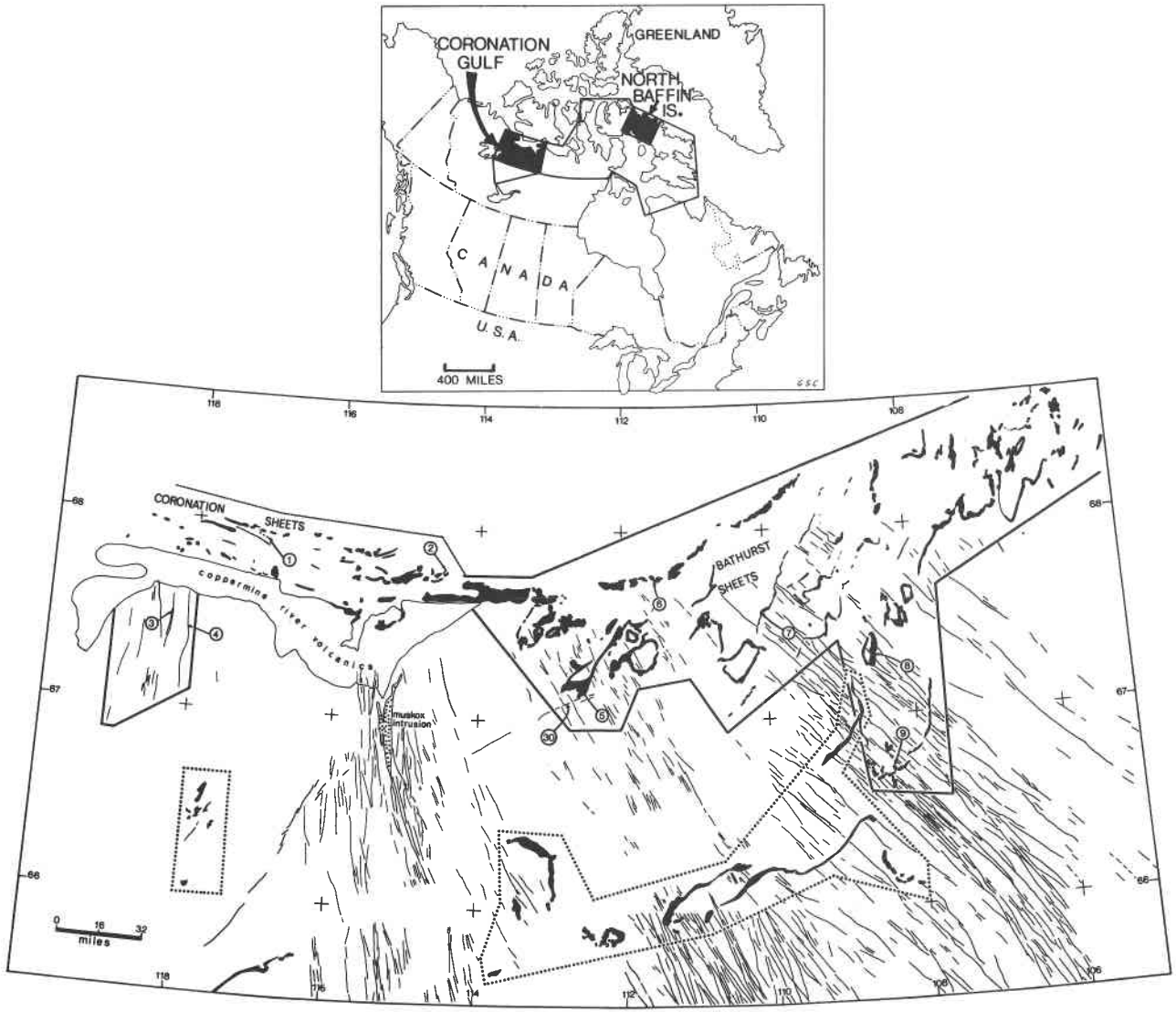


FIG. 1. The basic igneous rocks of the northern part of the District of Mackenzie. Franklin diabase is enclosed in solid or dashed line. The unenclosed bodies, which include the NW dike swarm, the Coppermine River Volcanics, and the Muskox intrusion, are all of Mackenzie age. Pre-Mackenzie diabase intrusions lie within areas outlined by dots. The index map above shows the general extent of Franklin diabases discussed in this paper (inside heavy line) with areas of more intensive study shown in solid black.

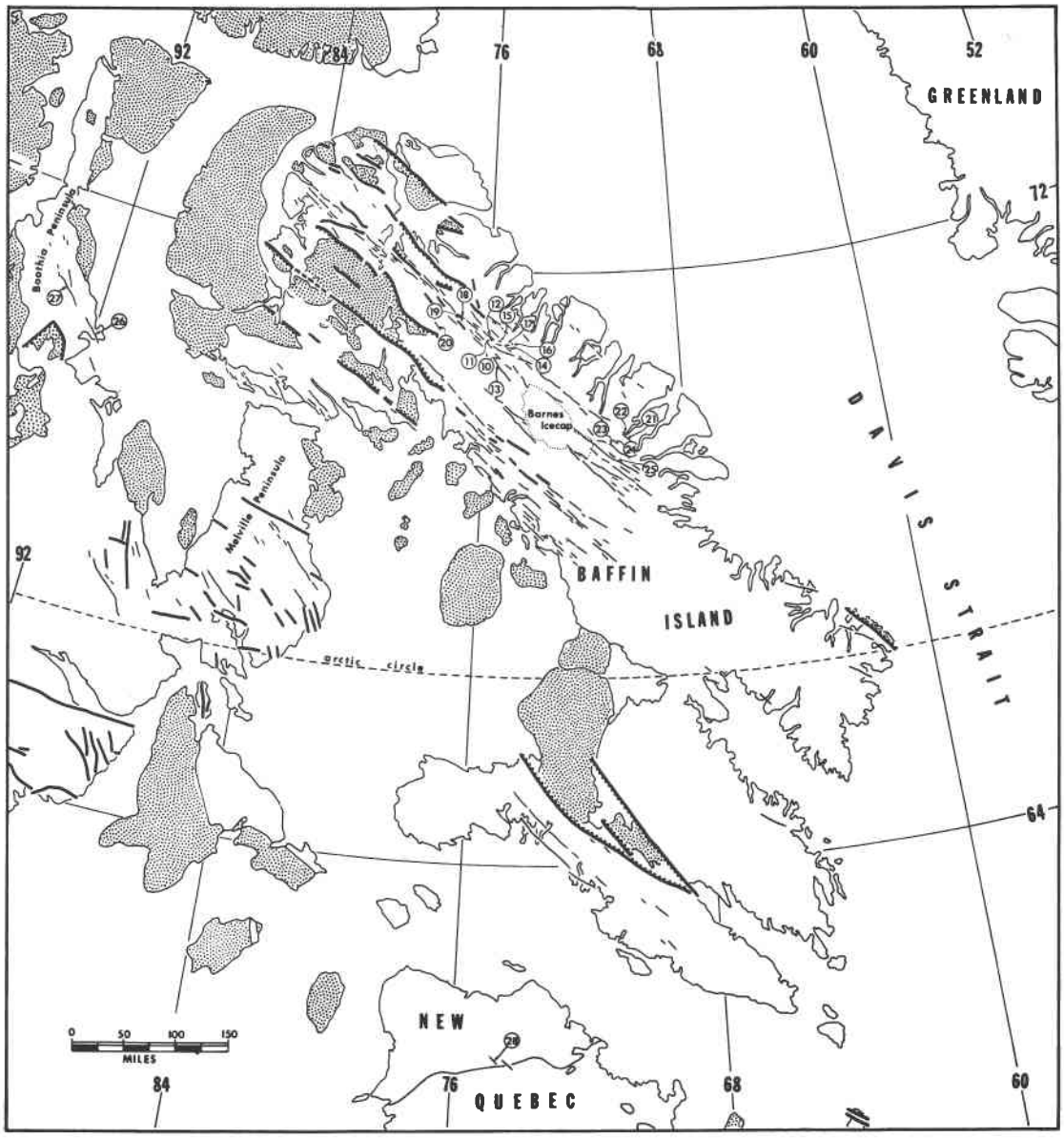


FIG. 2. Baffin dikes and the dikes of Boothia Peninsula and New Quebec. Paleomagnetic sampling sites are numbered. Heavy lines are faults, chiefly with normal displacement. Ticks indicate down-drop side where known. Stippled areas indicate Phanerozoic cover rocks.

from 655 to 695 (mean 674) were obtained by her, on biotite from baked country rock, on two whole-rock samples of chilled diabase, and on hornblende extracted from coarse-grained diabase.

In many instances the material available from a particular body of rock was not ideal for age work, and the reliability of the results varies widely. Many of the determinations were made prior to 1964 when improved laboratory equipment was installed (R. K. Wanless personal communication), so subsequent results are generally the more reliable. The average of the post-1963 results is 623 m.y. (standard deviation $\sigma = 77$ m.y.). Within this group the reliability varies, due to geological causes, and the best ages are probably those obtained from diabase at chilled contacts

TABLE 1. Potassium–argon ages of Franklin rocks

G.S.C. (collectors no.)	Material analyzed	K weight %	$^{40}Ar/^{40}K$	Radiogenic argon %	Year analyzed	Age	Paleomagnetic site
			Coronation and Bathurst Sheets and Dikes				
64-56 (FD-199-59)	ch wr	0.54	0.047	73	1964	605 ± 110	
N.P. (FA-15-66)	hb	0.42	0.0451	86	1967	647 ± 32	1
63-49 (FA-220-62)	ch wr	0.73	0.0692	100	1963	920	30
N.P. (FA-18-66)	bi hb	2.91	0.0438	96	1967	631 ± 40	6
64-69 (RN-136)	c wr	0.30	0.0511	62	1964	718 ± 120	
63-47 (FA-218-62)	ch wr	0.56	0.0436	100	1963	630	
63-56 (HF-79-1959)	c wr	0.80	0.0557	100	1963	770	
63-57 (CD-31/59)	c wr	1.33	0.0499	100	1963	705	
			Baffin Dikes				
64-33 (63BE43-2)	ch wr	5.80	0.0445	96	1963	639 ± 25	
N.P. (R-31-40-1954)	ch wr	0.36	0.0290	16	1970	439 ± 80	
N.P. (R-31-4-1954)	c wr	0.71	0.0288	44	1970	437 ± 60	
N.P. (FA-6-68)	bi cr	6.76	0.0505	99	1969	710 ± 31	14
N.P. (FA-680148)	ch wr	0.33	0.0410	90	1970	595 ± 90	14
N.P. (JD-C90/2-65)	c wr	0.75	0.0387	90	1968	566 ± 71	
N.P. (JDM-90/1-68)	c wr	0.29	0.0620	74	1969	841 ± 124	
N.P. (JDB-385/3-68)	c wr	0.49	0.0338	76	1969	504 ± 73	
N.P. (FA-680207)	ch wr	0.32	0.0448	82	1970	642 ± 98	20
N P (JD-C267/3-65)	ch wr	0.23	0.0472	77	1967	672 ± 110	
N.P. (FA-680256)	ch wr	0.27	0.0395	78	1970	577 ± 92	25
N.P. (FA-680217)	ch wr	0.61	0.0371	87	1970	546 ± 74	21
			Other Franklin Intrusive and Extrusive Rocks				
63-50 (FA-221-62)	ch wr	0.90	0.0692	100	1963	830	
64-64 (FA-162-62)	ch wr	0.80	0.0429	78	1964	619 ± 85	29
64-65 (FA-164-62)	ch wr	1.30	0.0425	92	1964	614 ± 65	
N.P. (FA-20-66)	bi hb	4.70	0.0512	97	1967	718 ± 27	
65-59 (FA-650052)	c wr	0.30	0.0418	60	1965	606 ± 32	
N.P. (FA-13-66)	ch wr	0.77	0.0419	96	1967	607 ± 75	26
N.P. (FA-96-63)	bi hb	0.72	0.0340	61	1965	507 ± 85	28
N.P. (FA-4-65)	ch wr	0.85	0.0362	83	1966	534 ± 74	28
62-83 (TC-26L9B)	c wr	1.6	0.0441	100	1962	635	
62-84 (TC-26L9A)	c wr	1.1	0.0447	100	1962	640	

NOTE: In the first column N.P. denotes unpublished data, whereas numbers indicate prior publication in the annual Geological Survey papers devoted to K–Ar isotopic ages; numbers beginning with 63 being reported in Paper 64-17, those beginning 64 in Paper 65-17, etc.

In the second column (ch wr) means chilled whole rock, (c wr) medium- or coarse-grained whole rock, (bi hb) biotite–hornblende concentrate, (hb) hornblende concentrate, and (bi cr) biotite from country rock contact. The errors quoted with the ages are estimates of laboratory error ($P = 0.05$). R-31-40 and R-31-4 were run earlier (1963) and much older ages (915 and 1140) were reported. The reason for this discrepancy is not known and the data has not therefore been used in the analysis reported in the text.

(whole-rock), on biotite or hornblende concentrates from the diabase, and on biotite extracted from baked country rock. The mean of 10 chilled contacts is 601 m.y. ($\sigma = 79$) and of 6 mineral separates (including Leech's data) 646 m.y. ($\sigma = 70$). The mean whole-rock age is the lower, and this may reflect minor argon loss, perhaps related to devitrification of a once glassy matrix. The best estimate of the mean K–Ar age of the intrusives is therefore about 650 m.y. based on mineral separates. As minor argon loss is likely to have affected to some degree, even the mineral separates, this mean K–Ar age may be somewhat lower than the true mean age. The true mean age probably lies within the range 650 to 700 m.y., and we shall accept 675 m.y. as the best estimate.

Paleomagnetic Methods

A total of 135 oriented drill cores, 2.5 cm in diameter, were collected from 30 sites. Orientation was by sun compass, using the methods of Larochelle (1961). The measurement of natural remanent magnetization (n.r.m.) was made using spinner magnetometers of commercial manufacture. Sites 1 to 9 and 26 to 30 were processed at the University of Alberta, Edmonton, and sites 10 to 25 (and all coercivity measurements) at the Earth Physics Branch, Department of Energy, Mines and Resources, Ottawa. Two specimens were cut from each core, and in the analysis average values for each core were used as the basic datum. The statistics are given in Table 2.

Alternating field (a.f.) demagnetization was carried out in order to remove unwanted secon-

TABLE 2. Directions and intensities of remanent magnetization before (n.r.m.) and after magnetic cleaning

Site no. (location)	N	Intensity	D, I	k	α_{95}°*	Pole
		Coronation and Bathurst Sheets and Dikes				
1. n.r.m.(118.3, 67.5)	5	4.30	random	—	—	—
1. 200 0e A.F. Field	4	0.31	080, −01	189	5	161, 03S
2. n.r.m.(111.6, 67.8)	4	8.49	078, +16	147	10	—
2. 100 0e	4	7.91	079, −10	217	5	170, 01N
3. n.r.m.(118.3, 67.5)	5	12.71	272, +63	98	6	—
3. 200 0e	5	4.88	262, +33	569	2	166, 13N
4. n.r.m.(108.0, 67.4)	5	12.00	264, +60	76	7	—
4. 150 0e	5	3.93	264, +43	124	6	177, 21N
5. n.r.m.(112.6, 67.2)	4	8.35	random	—	—	—
5. 250 0e	3	0.54	264, +14	28	13	166, 04N
6. n.r.m.(111.6, 67.8)	4	5.25	258, +68	28	13	—
6. 250 0e	4	0.73	266, +23	319	4	167, 10N
7. n.r.m.(110.0, 67.6)	4	10.92	274, +66	201	5	—
7. 250 0e	4	1.97	272, +43	1018	2	168, 24N
8. n.r.m.(108.6, 67.3)	4	16.79	274, +37	324	4	—
8. 250 0e	4	3.76	270, +20	664	3	165, 10N
9. n.r.m.(108.4, 66.7)	4	15.07	331, +65	7	26	—
9. 250 0e	3	1.34	280, +48	138	6	165, 30N
		Baffin Dikes				
10. n.r.m.(075.7, 70.9)	6	17.85	262, +25	4	42	—
10. 300 0e	6	0.85	273, −05	68	8	191, 01S
11. n.r.m.(076.0, 71.0)	6	6.00	272, +29	18	16	—
11. 500 0e	6	0.09	277, −21	8	25	184, 08S*
12. n.r.m.(076.0, 71.0)	7	8.30	305, +85	5	29	—
12. 300 0e	7	0.27	131, +72	28	12	138, 43S*
13. n.r.m.(075.8, 70.8)	7	8.54	327, +24	6	27	—
13. 200 0e	7	4.46	307, −24	133	5	155, 01S
14. n.r.m.(075.3, 71.0)	7	9.33	129, +77	12	18	—
14. 300 0e	7	0.36	119, +49	32	12	158, 20S*
15. n.r.m.(075.5, 71.0)	7	14.40	023, +81	20	14	—
15. 200 0e	7	0.92	109, +82	25	12	137, 62S*
16. n.r.m.(075.3, 71.0)	7	11.78	078, +75	19	14	—
16. 200 0e	7	0.42	099, +68	11	19	165, 45S*
17. n.r.m.(075.3, 71.0)	6	9.26	147, +76	8	26	—
17. 300 0e	6	1.12	118, +56	87	7	157, 26S*
18. n.r.m.(077.3, 71.4)	5	8.11	280, +42	3	51	—
18. 300 0e	5	0.68	296, −26	64	12	164, 05S
19. n.r.m.(077.3, 71.4)	5	5.20	292, +36	4	41	—
19. 500 0e	5	0.16	298, −26	200	5	162, 04S
20. n.r.m.(078.2, 71.2)	6	7.01	198, +45	4	42	—
20. 200 0e	6	1.56	301, −07	78	8	161, 06N
21. n.r.m.(070.3, 69.8)	6	10.08	315, +64	44	10	—
21. 400 0e	6	0.33	303, −22	24	14	165, 01S
22. n.r.m.(070.5, 69.9)	6	13.63	066, +72	22	15	—
22. 300 0e	6	0.90	105, +49	83	7	175, 23S*
23. n.r.m.(071.2, 69.7)	5	17.97	307, +35	14	21	—
23. 400 0e	5	0.49	288, −05	41	15	181, 04N
24. n.r.m.(070.9, 69.8)	5	8.80	285, +20	9	26	—
24. 400 0e	5	0.28	302, −13	9	26	167, 04N
25. n.r.m.(070.9, 69.5)	5	16.03	321, +48	206	5	—
25. 300 0e	5	4.12	313, +29	704	3	162, 29N
		Boothia and Other Dikes				
26. n.r.m.(091.9, 70.2)	5	3.91	128, +31	7	24	—
26. 250 0e	5	0.47	115, −13	271	4	157, 15N
27. n.r.m.(093.8, 70.8)	5	4.12	107, +37	9	21	—
27. 250 0e	5	0.48	112, −12	31	11	157, 13N
28. n.r.m.(074.7, 61.4)	6	11.28	112, +62	12	16	—
28. 250 0e	6	1.24	120, −12	333	3	171, 19N
29. n.r.m.(117.0, 54.5)	5	11.21	random	—	—	—
29. 250 0e	4	0.84	278, +20	80	8	153, 13N
30. n.r.m.(112.8, 67.1)	5	9.34	282, +63	7	24	—
30. 200 0e	3	1.03	263, −03	26	16	163, 04S

NOTE: The site location is given by longitude west and latitude north. N is the number of cores (two cylinders per core). The geometrical mean intensity for each site is given in c.g.s. units/cm^3 $\times 10^{-4}$ D,I are the declination and inclination of the site mean direction before (n.r.m.) and after cleaning in the alternating magnetic fields (peak) indicated. k and α_{95}° are Fisher's estimate of precision and cone of confidence at $P = 0.05$ (Fisher 1953). The pole chosen is that which falls in the Pacific (Fig. 5) and is prescribed in degrees of longitude east. Baffin Group B poles are indicated by an asterisk.

TABLE 3. Properties of individual specimens. Intensities of natural (n.r.m.), anhysteretic (a.r.m.), and saturation isothermal (i.r.m.(s)) remanent magnetization in c.g.s. units/cm^3, and estimates of their coercivities (median destructive fields m.d.f.) in oersteds, are given for one specimen each site. Ferric and ferrous oxide content (in %) at the center, and from chilled contact zones where available, are given.

Site	n.r.m. $\times 10^{-4}$	m.d.f. n.r.m.	a.r.m. $\times 10^{-4}$	m.d.f. a.r.m.	i.r.m.(s) $\times 10^{0}$	m.d.f. i.r.m.(s)	Contact		Center	
							Fe_2O_3	FeO	Fe_2O_3	FeO
				Coronation and Bathurst Sheets and Dikes						
1	2.80	062	9.59	123	0.189	198	—	—	2.4	10.4
2	7.93	200	26.10	207	0.560	346	—	—	3.1	12.2
3	10.19	125	33.90	210	0.915	379	—	—	2.9	11.8
4	14.71	065	15.30	138	0.328	231	—	—	2.4	8.3
5	8.21	080	19.40	157	0.361	270	—	—	1.3	11.3
6	6.51	107	25.20	178	0.526	313	—	—	2.9	10.5
7	9.02	127	30.10	203	0.690	350	—	—	1.2	14.2
8	18.88	160	19.20	166	0.353	285	—	—	2.3	9.8
9	18.62	080	35.90	153	0.791	295	—	—	2.2	12.9
				Baffin Dikes						
10	14.45	117	26.60	153	0.710	278	3.4	12.0	3.9	9.7
11	4.86	140	9.93	161	0.248	305	2.3	8.3	2.2	7.2
12	13.57	053	26.20	100	0.528	248	—	—	3.7	8.8
13	13.52	187	33.60	202	0.830	453	4.5	9.6	4.6	10.2
14	5.70	132	16.90	193	0.431	370	3.0	11.8	4.3	9.3
15	17.58	040	24.70	101	0.475	234	—	—	5.5	9.3
16	9.07	067	17.50	098	0.301	230	—	—	4.6	10.0
17	8.38	114	18.10	177	0.420	324	—	—	3.6	8.2
18	5.85	055	19.10	170	0.529	348	—	—	4.1	10.4
19	4.81	216	17.50	150	0.369	280	2.9	11.7	4.6	9.5
20	7.65	041	19.80	131	0.441	252	4.1	9.2	4.5	9.6
21	9.10	077	17.30	162	0.441	343	4.8	11.2	4.5	10.0
22	17.09	040	35.70	114	0.475	260	—	—	3.7	10.7
23	20.09	077	21.40	090	0.366	221	—	—	3.8	11.1
24	52.87	159	13.10	180	0.346	352	—	—	2.6	7.8
25	19.90	168	30.20	200	0.700	338	2.0	10.8	2.9	9.7
				Boothia and Other Dikes						
26	2.24	055	6.90	132	0.172	230	0.7	10.4	2.2	8.3
27	3.14	107	12.10	170	0.259	283	—	—	0.7	6.9
28	18.84	068	27.20	155	0.652	263	3.4	11.3	2.0	12.8
29	14.98	055	35.10	137	0.828	273	3.9	14.7	4.4	15.1
30	11.82	053	26.50	144	0.525	267	3.8	12.3	3.3	12.1

NOTE: The ferric and ferrous iron determinations were carried out by the classical analytical method (Pratt) in the Analytical Section of the Geological Survey of Canada (Analyst: G. Vickers). The a.r.m. is that acquired in an alternating field of 2500 0e and a bias steady field of 0.3 0e. The i.r.m.(s) is that induced in 11 000 0e.

dary components. The demagnetizing field was chosen by standard procedures (As and Zijdeveld 1958; Irving *et al.* 1961). Six specimens did not respond to a.f. demagnetization; their intensities were several times greater than those in other specimens from the same sites, and their magnetization is attributed to lightning. The results from these specimens (4 % of the data) were discarded. Coercivity of remanence was measured by determining the median point of the coercive force spectrum; that is, the strength of the field (the median destructive field or m.d.f.) required to reduce the intensity by one-half. Three estimates were made; the coercivities of n.r.m.; the anhysteretic remanent magnetization (a.r.m.) using a.f. methods; and the coercivity of saturation isothermal remanent magnetization (i.r.m.(s)) using d.c. methods. Because of the complicated geometry of the primary and secondary components, the estimates of coercivity of n.r.m. are crude and the coercivities of a.r.m. and i.r.m.(s) are the most useful for comparative purposes. Individual results are listed in Table 3 and the averages in Table 4.

Oxidation State

For comparison with the magnetic properties the percentages of iron oxides in the intrusions are given in Table 3. These were obtained from samples from centers of intrusions, and also, where possible, from chilled contacts. The

TABLE 4. Comparison of mean coercivities and iron oxide ratios

	m.d.f. n.r.m.	m.d.f. a.r.m.	m.d.f. i.r.m.(s)	i.r.m.(s)	$\frac{FeO}{Fe_2O_3}$
Coronation and Bathurst intrusives (9)	111	170	296	0.524	4.4
Boothia and other dikes (5)	068	148	263	0.487	5.4
Baffin dikes group *A* (10)	124	160	317	0.498	2.6
Baffin dikes group *B* (6)	074	135	278	0.438	2.3
All Baffin dikes (16)	105	151	302	0.475	2.5
Grand average (30)	101	156	294	0.492	3.6

NOTE: The number of sites represented (one specimen per site) is given in the first column. Other abbreviations as in Table 3.

former are used below, because values were available for all sites, and because they are more appropriate for comparison with the paleomagnetic results which were also obtained from samples derived chiefly from the centers of bodies. The mean oxidation ratios FeO/Fe_2O_3 (high values indicates low oxidation state) are listed in Table 4.

Results

The n.r.m. directions of the Coronation and Bathurst intrusions showed variable dispersion. Two sites were internally random (Table 2). After a.f. demagnetization between 100 to 250 Oe (peak) the dispersion within-sites decreased and the site mean directions became grouped about an east–west axis with low inclination. There are 7 sites with westerly declination (referred to as normal polarity) and 2 sites with an easterly declination (reversed polarity). The stable component remaining after a.f. cleaning is much smaller than the initial n.r.m. intensity; the average ratio of the mean site intensities before and after cleaning is 0.24. The oxidation state is low (Table 4).

The Boothia and other dikes have coercivities which are generally lower than those of the Coronation and Bathurst intrusions. Alternating field cleaning greatly improved the within- and the between-site precision, and the directions grouped together about an east–west axis. Two sites were normal and 3 were reversed. The average of the ratios of intensities before and after cleaning is 0.10. The oxidation state is low.

Most of the Baffin dikes (10 dikes referred to as group *A*) responded well to a.f. demagnetization, the mean site directions moving 40° or more away from the direction of the present field. Nine sites were normal and one reversed. The ratios of intensities before and after cleaning averaged 0.13.

The paleomagnetic pole positions for Bathurst and Coronation intrusions, Boothia and other dikes, and the group *A* dikes of Baffin Island are shown as solid dots on Fig. 4. Their mean pole is 167° E, 8° N (K-32.3, $\alpha_{95} = 5°$, N-24, 19 normal and 5 reversed sites). The dispersion of these poles is comparable to that expected (K-26) from the dispersion of the present field in low latitudes (Cox 1962) and may therefore be attributed to the secular variation.

The remaining 6 Baffin dikes, called group *B*, behaved differently. Initially, the inclinations were steep, but, after partial demagnetization in fields of 200 to 300 Oe, the directions became spread out towards the east, and between-site dispersion increased. Cleaning in higher fields increased the scatter still further, and at 500 Oe all the *B* sites were internally random. Group *B* poles are shown as open circles on Fig. 4.

All the Baffin dikes have low FeO/Fe_2O_3 ratios (Table 4). The coercivities of Group *A* are comparable to those of the other Franklin diabases. The coercivities of Group *B*, as a whole, tend to be lower, but there is no clear-cut separation. However, within group *B*, the coercivities decrease with increasing angular divergence from the mean pole of the other Franklin diabases (Table 3); for instance, site 15, which yielded a pole farthest from the mean position, has a coercivity of a.r.m. of only

101 Oe (Table 3). The low stability of the Group *B* Baffin dikes is also shown by the ratios of intensities before and after cleaning, which are very low, averaging 0.06.

In order to investigate the reason for the high oxidation state of the Baffin dikes one polished section for each was examined. The percentage of opaque minerals ranges from 2.5 to 12. Ilmenite is the dominant type. It occurs as sub-skeletal grains, as widmanstatten-like lamellae inside larger composite grains, and as variously shaped intergrowths near the edges of these grains. The intersticies between the ilmenite lamellae are filled with fine-grained minerals which are difficult to identify; ilmenite, magnetite, and, in some sections, goethite, are present, and examination in plane light, under high magnification, revealed the presence of a pale blue mineral. The fine grain-size makes identification uncertain, but this blue mineral is thought to be maghemite (R. McLeod, personal communication). McLeod (1970) has also identified maghemite in iron formation from the Mary River area of northern Baffin Island. Sulfides are present in small quantities, the largest amount being 1 % chalcopyrite at site 20. At all 16 Baffin sites chalcopyrite and pyrite are about equally abundant. There are traces of bornite in four sections, and chalcocite in one. The opaque mineral assemblages in group *A* and *B* dikes were indistinguishable. The abundant ilmenite lamellae are presumably due to high-temperature exsolution from titanomagnetite during initial cooling. The presence of goethite, and possibly of maghemite also, suggests that a secondary low-temperature oxidation has also occurred, either during the last stages of cooling, or during a period of deep weathering before the deposition of Ordovician sediments which may once have covered all of Baffin Island. Whatever the nature and date of this process, it evidently was not responsible for the magnetization of either the group *A* dikes (since their directions agree with those on the mainland) or the group *B* dikes (since both group *A* and *B* dikes are oxidized). Regarding the origin of group *B* magnetizations, it is notable that a line can be drawn separating group *B* dikes from group *A*, which places them on the side of the swarm nearest to Davis Strait, very close to what was probably the early Tertiary rift zone between Baffin and Greenland (Fig. 3). It is therefore possible that a secondary magnetization was acquired by the group *B* dikes at a time of igneous activity and mild reheating in the rift zone during the latest (early Tertiary) opening of Davis Strait.

Geochronological Aspects

Detailed paleomagnetic observations have shown that Phanerozoic rocks of the same geological period from the same continental block have the same pole positions, whereas rocks within the same blocks but belonging to younger or older periods generally have different poles. These apparent polar changes are due to movements of the continental blocks relative to the earth's field (by continental drift for example) and are caused by dynamic processes occurring in the earth's mantle. Superimposed on this pattern of apparent polar wandering are reversals in polarity of the geomagnetic field. Reversals, like the field itself, originate in the earth's core. They occur with a frequency which is known to vary by about 500, from 0.1 m.y. (Ninkovitch *et al.* 1966) to 50 m.y. (Irving and Parry 1963). During some periods the field has not reversed (the Permian for example), whereas at other times it has done so frequently (in the Pliocene for example). Thus, rocks of a given age are characterized, not only by a corresponding pole, but also, to a degree, by the presence or absence of reversals. An excellent illustration of this principle is found in Larochelle (1968, Fig. 4). This is of some chronostratigraphic interest, because eventually it may be possible to divide the Phanerozoic and Proterozoic into intervals of time defined by the paleomagnetic characteristics of the rocks formed during these eras. There are two terms that are useful in this regard, namely, *polarity intervals* which are periods of time during which the same polarity was maintained, and *magnetic intervals* (Irving and Robertson 1969) which are periods of time during which the rocks of a particular area indicate a constant pole position and the earth's polarity has a particular reversal frequency. Since pole positions reflect the movements of continents, a sequence based on magnetic intervals may be related to major

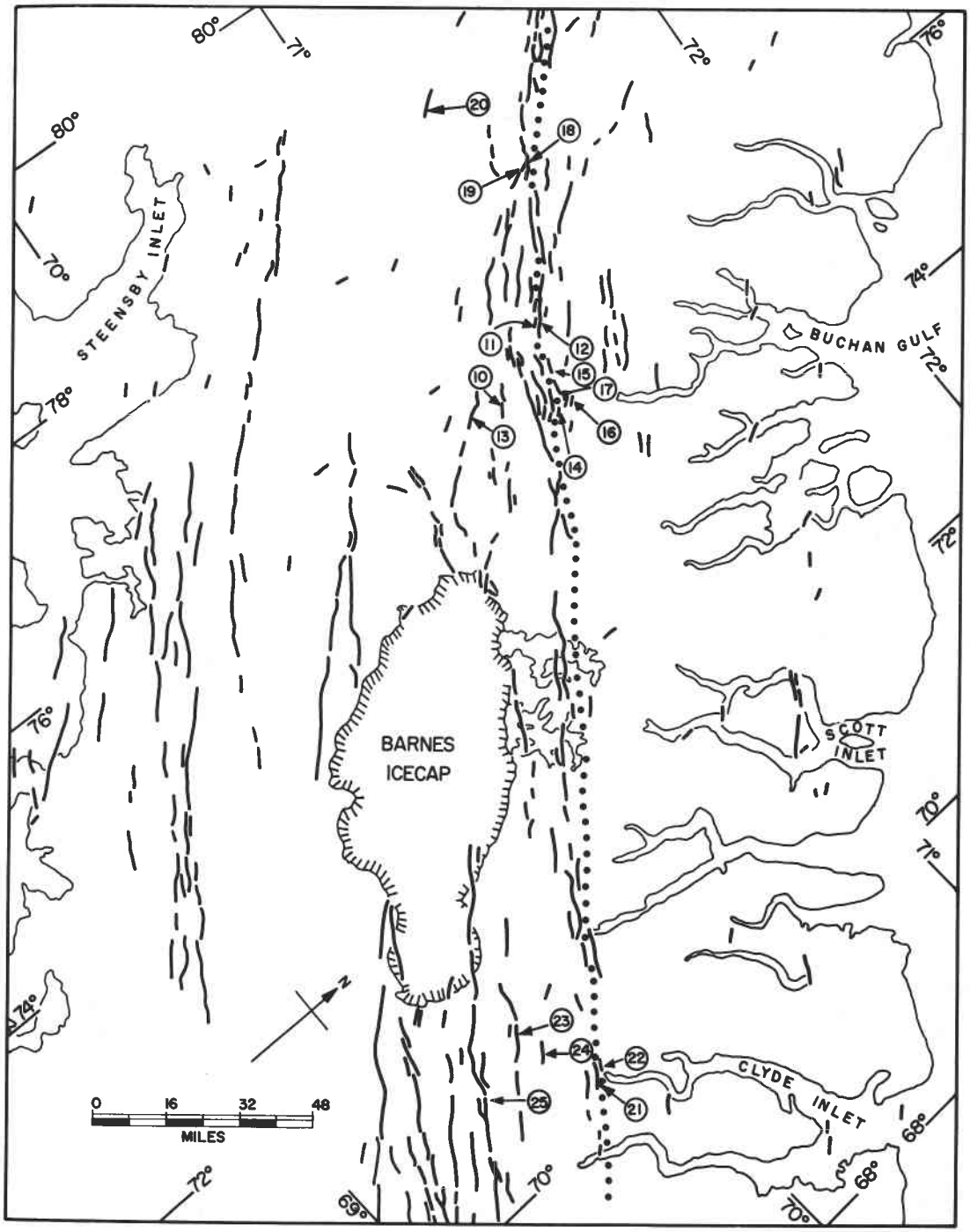


FIG. 3. Detailed distribution of Baffin dikes. Group *B* dikes are to the east of the dotted line; Group *A* dikes are to the west.

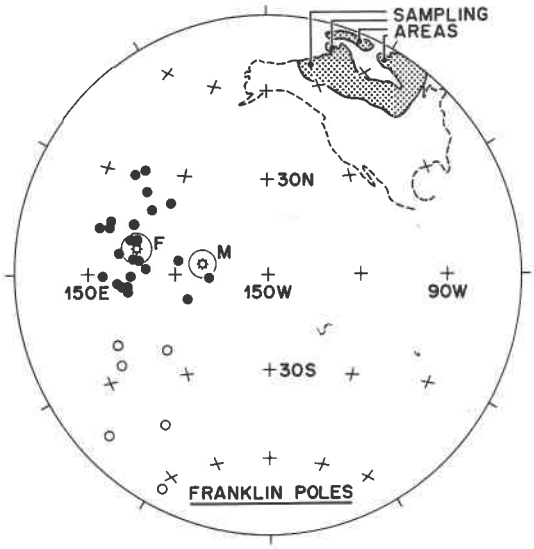


FIG. 4. Franklin poles. The solid dots are the poles (both north and south) for the group *A* dikes of Baffin, the Coronation and Bathurst intrusives, and the Boothia and other dikes. The open circles are poles (south) from Baffin Group *B* sites which are thought to possess a component of magnetization induced at the time of opening of Davis Strait. *F* and *M* are the mean pole for the Franklin (675 m.y.) and Mackenzie (1200 m.y. Fahrig and Jones 1969) Intervals with 95 % Fisher circles of confidence.

diastrophism, and hence is likely to have an ordered relationship to chronostratigraphy. In this paper the term magnetic interval is used to denote intervals of time during which in the type area rocks exhibit a number of characteristic paleomagnetic properties; these are mean pole position, reversal frequency, and eventually perhaps dispersion of the field, in that order of priority. There are many advantages to having such a paleomagnetic sequence for the Precambrian (for example in order to provide an independent check on lithostratigraphic sequences) and as a first step, the definition of two Precambrian magnetic intervals is proposed.

In earlier papers (Robertson 1969; Fahrig and Jones 1969) geological and paleomagnetic evidence was given for suggesting that about 1200 m.y. ago (Helikian) there were very wide spread basic igneous intrusive and extrusive activities that affected virtually all of the western and central Canadian Shield. These activities include the formation of the Muskox Intrusion, the outpouring of the Coppermine lavas, and the formation of NW trending dike swarms of the District of Mackenzie and the Sudbury area. During these activities, the geomagnetic field for the western and central shield (recorded paleomagnetically) had a characteristic direction (defined by the pole *M* of Fig. 4) and there were no reversals. The period of time during which the field in this area had this configuration, and during which no reversals have been observed, may be referred to as the Mackenzie Magnetic Interval (or simply the Mackenzie Interval). Its mean age is 1200 m.y. During the intrusion of the Franklin diabase the field was different, and one may speak therefore of a Franklin Magnetic Interval, with a mean age of 675 m.y., and a field configuration for rocks in the northern shield defined by pole *F* (Fig. 4) and during which at least one field reversal occurred. The Mackenzie and Franklin poles, although significantly different, are not a great distance apart. There is no confusion, however, since between them there is a very large polar loop which swings toward North America. This effect, discovered by DuBois (1962), has now been documented in some detail and these new results will be reported later. They are noted here because the mapping of such large field variations should make it possible to define a set of Magnetic Intervals between the Mackenzie and Franklin Intervals.

Hadrynian Paleolatitude and Climate

The inclinations of the Franklin rocks are low, indicating that they were formed in low latitude (Fig. 5), and it is of interest to review the geological evidence of climate conditions. The 1000 m sedimentary sequence intruded by Coronation sheets, and which is probably only a little older than them, has been correlated with a gypsum and stromatolitic carbonate-bearing sequence on Amundsen Gulf (Fraser 1960). The Natkusiak Volcanics of Victoria Island are Hadrynian in age (Thorsteinsson and Tozer 1962). They overlie the Shaler Group disconformably, and both the Shaler and Natkusiak are intruded by diabase that does not cut Cambrian strata and has yielded K–Ar ages in the Franklin range (Table 1, nos. 26L9A and B). The diabase of Victoria

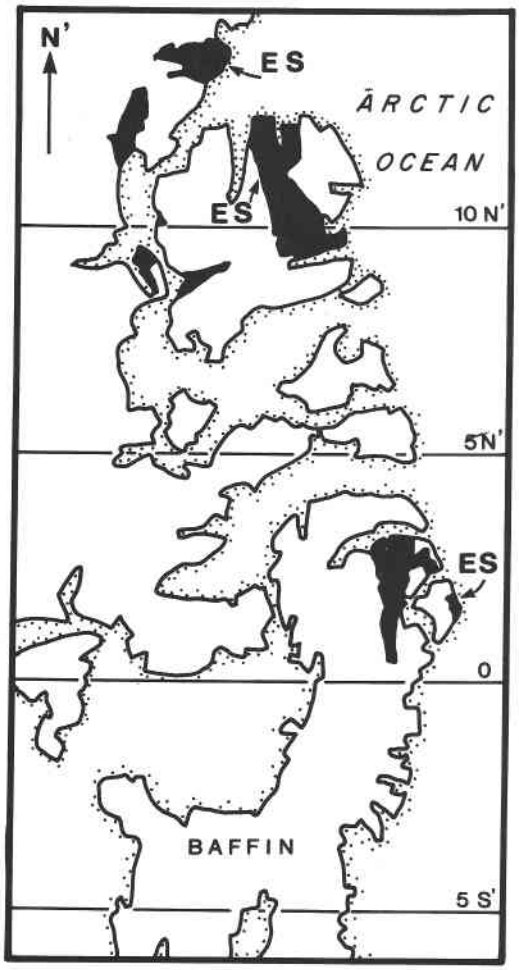


FIG. 5. Paleolatitude of the northern Canadian Shield during the Franklin Interval calculated from the pole *F* of Fig. 5. Outcrop of Hadrynian (with possibly some late Helikian) beds is shaded. Occurrences of evaporites and stromatolites are indicated by the letters *E* and *S*.

Island as well as the Natkusiak Formation are therefore probably Hadrynian. The underlying Shaler Group may also be late Proterozoic, and it contains beds of gypsum, anhydrite, and stromatolitic carbonate. In northern Baffin Island, unmetamorphosed, flat-lying to gently-folded, volcanic and sedimentary formations of the Uluksan and Eqalulik Groups (Blackadar 1970) are intruded by Franklin dikes that do not intrude overlying lower Paleozoic strata. Jackson (1969) has correlated a sedimentary sequence containing gypsum and stromatolites in the area southeast of Milne Inlet and on Bylot Island, with the Uluksan and Eqalulik Groups. These gypsum-bearing sequences on northern Baffin Island on Victoria Island are classified as Hadrynian on the newest geological map of Canada (Douglas 1969), whereas rocks of similar lithology in Amundsen and Coronation Gulf area are shown as Neohelikian.

Thus, thick, late Proterozoic sequences, occurring as erosional remnants in an elongated area from Amundsen Gulf to northern Baffin Island, contain evidence of warm depositing conditions (gypsum, anhydrite, stromatolites). The ages of these sedimentary sequences have not yet been established, and they must be older than the intrusives on which the paleomagnetic evidence is based, but it seems likely that they are of comparable age. The low latitudes calculated from paleomagnetism is in good agreement with the geological evidence of climatic conditions.

Tectonic Speculations

Diabase dike swarms indicate crustal tension and it is of interest to relate them to other regional tectonic features of similar ages. There are two such features sub-parallel to the Baffin dikes: a spectacular NW trending fault system, and the trend of the Island itself (Fig. 2). The movement on most of the faults is normal, and has resulted in the development of a series of grabens. Trettin (1969) argues that in NW Baffin, one such zone (the Central Border Fault zone) was an important hinge-line during Helikian sedimentation, and, along this zone, and prior to lower Paleozoic sedimentation, crystalline basement rocks were elevated to the level of Helikian strata. Because isopachs of the lowest Paleozoic strata are partly parallel with this zone, Trettin suggested that the zone was weakly active during lower Paleozoic sedimentation and was reactivated in post-Silurian time. However, it is possible that the Helikian sedimentation to which Trettin refers, is actually Hadrynian, so that the formation of the NW trending Baffin dikes and the NW trending graben zones may be roughly concordant and due to the same regional tension. This introduces the possibility that what is now Davis Strait was a rift zone in very late Precambrian time and may therefore contain Paleozoic strata. The very much later Cenozoic separation of Baffin Island and Greenland, indicates ten-

sion similar in direction to that indicated by the late Precambrian dikes and the normal-fault systems of Baffin Island. Thus, NE–SW tension may have occurred several times.

It is also interesting to note that these late Precambrian paleomagnetic results provide us with a means of studying the Paleozoic history of the Atlantic. The deep-water deposits of the Caledonian geosyncline of NW Europe and their equivalents in eastern North America indicate the presence of an ocean in the early Paleozoic, which closed in the middle Paleozoic during the Caledonian orogeny, and reopened in the Mesozoic to form the Atlantic (Wilson 1966). At first sight it would appear that the way to test such ideas is to carry out paleomagnetic studies on material from within the orogenic belts themselves, but such rocks have generally been extensively remagnetized (Chamalaun and Creer 1965; Roy *et al.* 1967) and any hope of getting a sufficient quantity of hard information seems remote. A procedure, which is however experimentally accessible, is to compare Proterozoic polar sequences from the Canadian and Baltic Shields, which have formed the marginal cratons, and which have supposedly been carried along in this to and fro motion. If a wide early Paleozoic ocean did not exist, then equivalent poles will agree when the continents are restored to their late Paleozoic positions. Now that paleomagnetic data is forthcoming from the Baltic Shield (Neuvonen and Grundstrom 1969) it should soon be possible to make a decisive test. Results in the age range between the Franklin and Mackenzie Intervals, will be of particular interest, because if the current interpretation according to plate tectonics is incorrect then the Franklin–Mackenzie polar loop should be identifiable in both shields.

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